



DEALER BEST PRACTICE SERIES

Component Rebuild
Customer Reports

Application Maintenance **Component Life
Management** **Component
Rebuild** MARC
Management

Component Rebuild Customer Reports 0

1.0 Introduction 1

2.0 Best Practice Description..... 1

3.0 Implementation Steps 2

4.0 Benefits..... 3

5.0 Resources Required 3

6.0 Supporting Attachments / References 3

7.0 Related Best Practices 3

8.0 Acknowledgements..... 3

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

In order to achieve mining customer cost per ton objectives, customers must manage the component life of their equipment fleet. The key success factors in managing component life are to minimize unplanned failures and to optimize planned component rebuild intervals.

Planned component rebuild intervals can be based on a number of factors including past experience, condition monitoring, fuel consumption, oil analysis, or even arbitrary target hours.

Minesite conditions can change over time and impact component life. Negative impacts can come from deeper pits, longer hauls, overloading, as well as maintenance and monitoring. Positive changes can come from continuous improvement initiatives such as contamination control, fluid cleanliness, application and payload management, and better maintenance and monitoring.

When a component is repaired or rebuilt in the dealer CRC, dealers are able to assess the condition of the component. Dealer-documentation of the component's condition can be valuable information for the customer. Some dealers have provided technical reports back to the customer after every component rebuild or repair. These reports may also offer recommendations and information the customer can use to better manage component life.

2.0 Best Practice Description

This Best Practice contains several standard reports from MCTS used by some dealers to provide information and recommendations to mining customers.

Failure Analysis Report

This report is generated any time a component has suffered a failure or unplanned rebuild or repair. The report includes:

- Component identification
- Hours, rebuild history (and total fuel consumption for engines)
- Any applicable supporting data: VIMS / SOS / Particle Count
- Root Cause of Failure identification
- Recommendations to avoid future failures

Component Rebuild Report

This report is generated for every planned before-failure component rebuild. The purpose of the report is to assess the condition of the component and potential remaining life. The report includes:

- Component identification
- Hours, rebuild history (total fuel consumption for engines)
- Any applicable supporting data: VIMS / SOS / Particle Count
- Technical assessment of internal wear.
- Recommendations to improve life or adjust PCR interval.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Component Rebuild Customer Report	DATE 3/18/2016	CHG NO 03	NUMBER 1106-4.5-1040
-----------------------------------	-------------------	-----------------	-------------------------

Fleet Component Life Summary

This is a management report usually generated once a year and presented to the customer. It provides an overview of average component life by model and by component. It also reports the percentage of rebuilds performed before-failure and after failure.

- Average component life by component and model.
- Average component life for planned before-failure rebuilds
- Average component life for after-failure rebuilds.
- Percent of rebuilds made before failure.
- Percent of rebuilds made after failure.

A sample report is attached. This example was generated using the Caterpillar Major Component Tracking System (MCTS).

Major Component Life Query

Main Store Dealer CodeR460Report Date03-APR-2003

Customer NameALLSales Model793B

Component TypeENGINEJob SiteALL

Job Site TypeALLLife Cycles Up toALL

Component S/N RangeALL

Close

Life	Avg. Component Life		Repair Type							Actual Sample Size
	Percent	Avg. Hours	Failure Repair		Wearout		PCR		Unknown	
	Percent	Avg. Hours	Percent	Avg. Hours	Percent	Avg. Hours	Percent	Avg. Hours	Percent	
1	100%	15,040	5%	8,510	12%	11,098	78%	16,140	2%	71
2	100%	12,108	22%	6,848	16%	10,910	40%	15,467	20%	54
3	100%	5,277	45%	4,182	5%	10	15%	10,320	35%	20
4	100%	0	0%	0	0%	0	0%	0	100%	4
5	100%	9,521	50%	9,521	0%	0	0%	0	50%	2

Caterpillar Confidential: Green

©Copyright 2001 Caterpillar Inc. / All Rights Reserved.

The component life information is also sometimes reported as a histogram bell curve graph.

3.0 Implementation Steps

1. Determine level of customer and dealer interest in CRC reports and specific needs.
2. Create standard report forms, per customer and dealer feedback.
3. Create a process to collect the information and create the report. Assign ownership.
4. Implement a component life tracking system. (See the Caterpillar **Major Component Tracking System** Best Practice under Site Management – Component Life)
5. Develop customer communication process for these reports with feedback.
6. Document Benefits.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Component Rebuild Customer Report	DATE 3/18/2016	CHG NO 03	NUMBER 1106-4.5-1040

7. Develop follow-up assessment 6-12 months after implementation to measure actual benefits.

4.0 Benefits

- Improves customer cost per ton and preference for Caterpillar equipment.
- Builds customer relations.
- Contributes to both dealer and customer Continuous Improvement initiatives.
- Controls cost and risks of MARC contracts.
- Increases components life.
- Reduces component rebuild cost.

5.0 Resources Required

Personnel time to create and manage report system and process.

6.0 Supporting Attachments / References

Dealers interested in using MCTS should contact:

Jon Duvall
MCTS System Administrator
Duvall_Jon_D@cat.com
+1 309 675-7115

<https://mcts.cat.com/>

7.0 Related Best Practices

1106-4.5-1036	Component Life Tracking a Valuable CRC Performance Indicator
0906-3.1-1020	Major Component Tracking System Used to Monitor and Manage Component Life

8.0 Acknowledgements

This Best Practice was written by:

Name: Dale Brehm
Title: Global Mining Product Support
E-mail: brehm_dale_e@cat.com
Phone: 309 –675-6325

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Component Rebuild Customer Report	DATE 3/18/2016	CHG NO 03	NUMBER 1106-4.5-1040